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AnalystLab Africa - Data Science Internship Programme

Week 5 Assignment

Data Preparation and Baseline Modelling Report

**Improving Patient Appointment Attendance and Healthcare Support Using Data and
AI**

HealthConnect Clinic Experience Lab

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Internship: Data Science

Batch: D

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1. Executive findings and Week 4 continuity

I continued the HealthConnect Data Science project rather than selecting a new dataset. Week 4 established the business problem, the attended-versus-no-show target, candidate predictors and the need for a time-aware evaluation. Week 5 implements those decisions, with explicit revisions where timing was insufficiently specified.

My objective was to establish a reliable baseline for appointment-level no-show risk, not a final operational system. On 799 later bookings, logistic regression achieved 60.83% accuracy and 0.6678 ROC-AUC. Its precision was 58.85% and recall 63.57%. These results suggest limited predictive signal in the supplied fictional data, not readiness for clinical use or proof that attendance will improve.

Decision	Week 5 implementation
Target retained	No-Show = 1; Attended = 0
Cancellations retained separately	263 excluded from binary analysis
Prediction timing refined	At booking, rather than unspecified pre-appointment timing
Timing-sensitive features deferred	No reminder activity or waiting time as inputs
Evaluation strengthened	Earlier completed outcomes vs later bookings; boundary cases purged

IBM methodology connection: I revisited Business Understanding and Analytical Approach, then applied Data Understanding, Preparation, Modelling and Evaluation. Deployment and operational feedback remain future stages. I used the Week 4 report sections 3-9 and the Week 5 Data Science track requirements as the basis for these decisions.

The source CSV, data dictionary and Week 4 documents provide context. The code and exported metric files provide the evidence for the numerical results in this report. Original source files remain unchanged, with a SHA-256 digest recorded in run_summary.json.

2. Preparation and data-quality decisions

I loaded 5,000 rows and 18 columns with 1,696 distinct patient IDs. I preserved the literal reminder category `None` using `keep_default_na=False`, explicitly converted blank numerical entries to missing values, and parsed month/day/year dates. No invalid parses, exact duplicate rows, duplicate appointment IDs, booking-lead mismatches or weekday mismatches were found in this run.

Observation	My action and remaining limitation
Distance: 90 missing (1.8%)	Retain rows; fit median imputation on training data only.
Waiting time: 60 missing (1.2%)	Exclude from predictors because booking-time availability is unverified.
737 Sunday appointments	Retain and flag the knowledge-base conflict; no invented date corrections.
1,343 IDs with age span >2 years	Do not derive longitudinal history from IDs; request clarification.
1,366 reminder channels = <code>None</code>	Retain as a valid category in source audit; reminders excluded from model.
History counts lack timestamps	Use supplied counts conditionally as pre-booking history; unresolved leakage risk.

I checked that previous no-shows never exceed previous appointments, numerical values are nonnegative where present, appointment dates do not precede booking dates and target labels are expected. The notebook stops if these checks fail. Converting invalid strings to missing values is not enough: the failure counts must also be inspected.

I did not remove statistically unusual but plausible distances automatically. I also did not correct conflicting patient ages or Sunday appointments without evidence. These decisions preserve traceability and avoid replacing uncertainty with fabricated values.

3. Cohort and chronological test design

I excluded 263 cancelled appointments from the initial binary cohort, leaving 4,737 records. This experiment is conditional on the final outcome being attended or no-show; because future cancellation is unknown at booking, the model is not yet valid as a full all-bookings workflow.

Partition	Records	Rule
Training	3,682	Appointment completed before 1 March 2026
Test	799	Booking on/after 1 March 2026
Purged boundary cases	256	Booked before cutoff, appointment on/after cutoff

The cutoff was specified as a calendar boundary, not selected to maximise results. Purging prevents training on outcomes that would not yet be known when later bookings are scored. Model-directed plots and feature interpretation use training data; test outcomes are reserved for final baseline evaluation. This corrects the earlier outline in which deeper exploration appeared before the split.

There are 579 patient IDs shared across the training and test partitions. That is consistent with a later-appointments use case involving returning patients, but does not demonstrate performance on unseen patients. Because IDs have inconsistent ages, grouped evaluation also requires care. No patient ID is a predictor.

The training no-show rate is 51.11%; the test rate is 48.44%. Training and test class distributions were not artificially balanced. The source ends in June 2026, so later bookings with appointments beyond the observed window may be underrepresented. This limits generalisation to all future bookings.

Processed cohort exports retain train/test/purged labels and the target for audit purposes. They are not already-imputed model matrices. Identifiers, the target and split labels must never be accidentally included in the predictor matrix.

4. Visual evidence supporting preparation

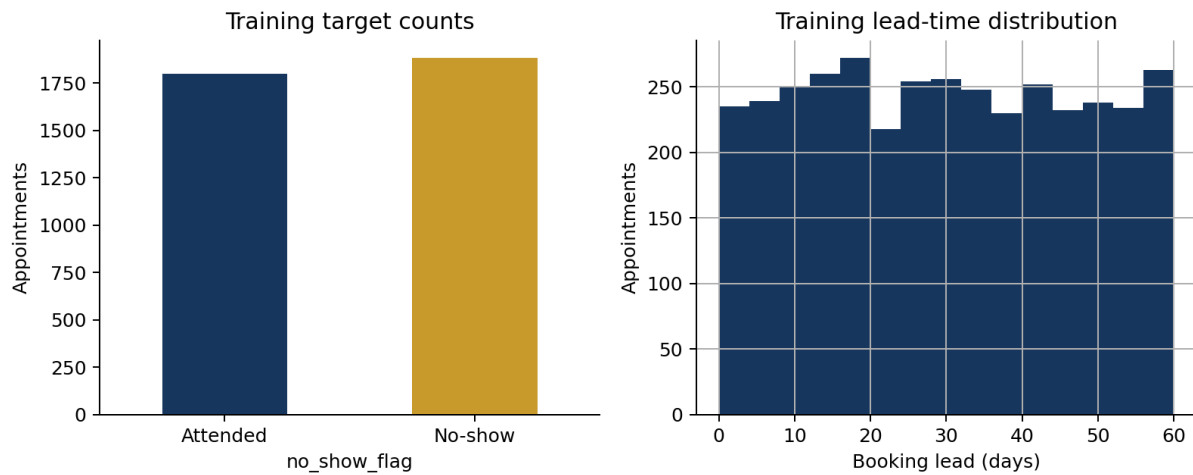


Figure 1. Both target classes have substantial training support. I kept their natural frequency and did not oversample. Lead-time variation supports retaining scheduling information; I used $\log(1 + \text{lead days})$ as a pre-specified nonlinear feature rather than tuning transformations on the test set. Improvement from this transformation alone has not been established.

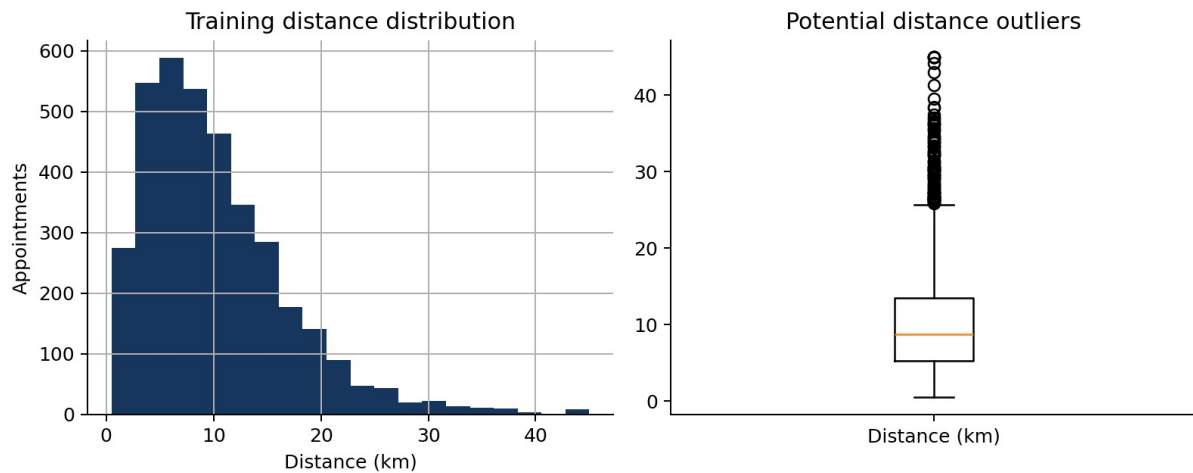


Figure 2. Distance distributions and boxplot flags help identify unusual values. They do not alone establish errors. I retained valid distances, used a training-only median for missing entries and standardised numerical inputs for regularised logistic regression. Week 6 can test robustness to outlier treatment within training-only validation.

5. Feature engineering and selection

Engineered feature	Definition and expected value
prior_no_show_rate	Previous no-shows / previous appointments; summarises attendance history. Zero when denominator is zero.
no_prior_history	1 when previous appointments = 0; distinguishes unknown history from observed perfect attendance.
is_weekend	1 for Saturday/Sunday scheduled appointments; captures a calendar pattern known at booking.
log_lead_days	$\log(1 + \text{booking lead days})$; allows a nonlinear lead-time relationship and retains same-day bookings.

My nine predictors are age, distance_to_clinic_km, previous_appointments, prior_no_show_rate, no_prior_history, is_weekend, log_lead_days, appointment_type and appointment_time. I omit raw lead days, raw previous-no-show count and redundant age/day categories from this simple specification. These are modelling choices, not proof that omitted features are universally useless.

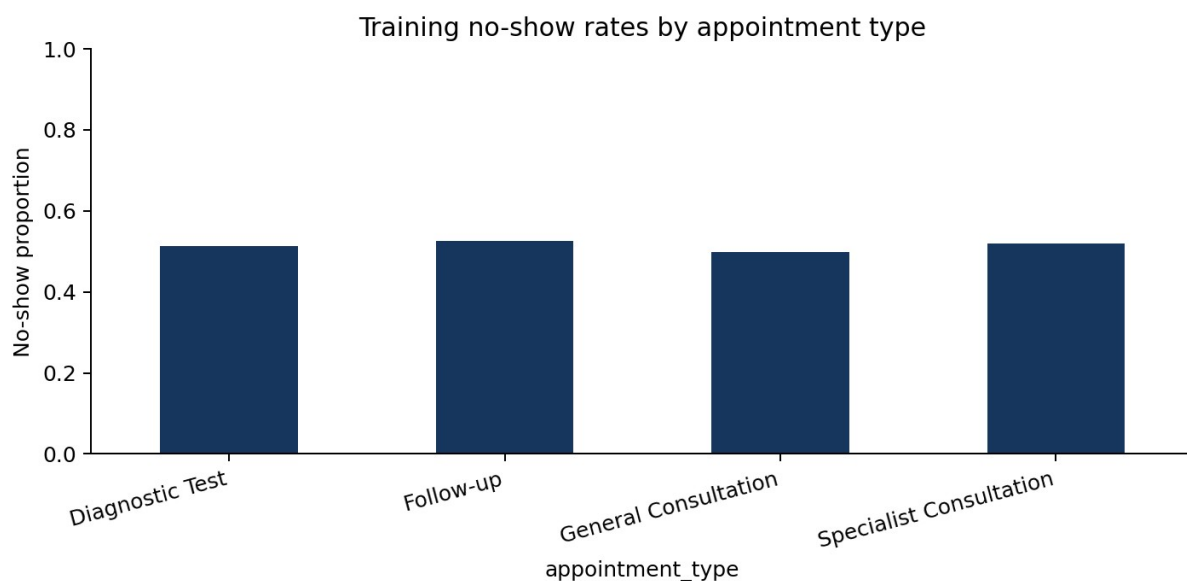


Figure 3. Appointment-type differences on training records motivate retaining the category through one-hot encoding. Counts and exact rates are exported in training_type_rates.csv. These are associations, not causal explanations or guarantees of future predictive value.

Gender is used for subgroup evaluation, not prediction. Age remains an input but requires review. Removing one protected characteristic does not guarantee fairness because other variables may encode related information. Estimated waiting time and reminders remain excluded until timing is verified.

6. Model development and initial evaluation

I fitted numerical median imputation and StandardScaler, plus categorical most-frequent imputation and OneHotEncoder(handle_unknown="ignore"), inside a ColumnTransformer and Pipeline. All learned preprocessing uses only the 3,682 training records. I fitted a most-frequent DummyClassifier and logistic regression with C=1 and max_iter=2000. No test-driven tuning or threshold optimisation was performed.

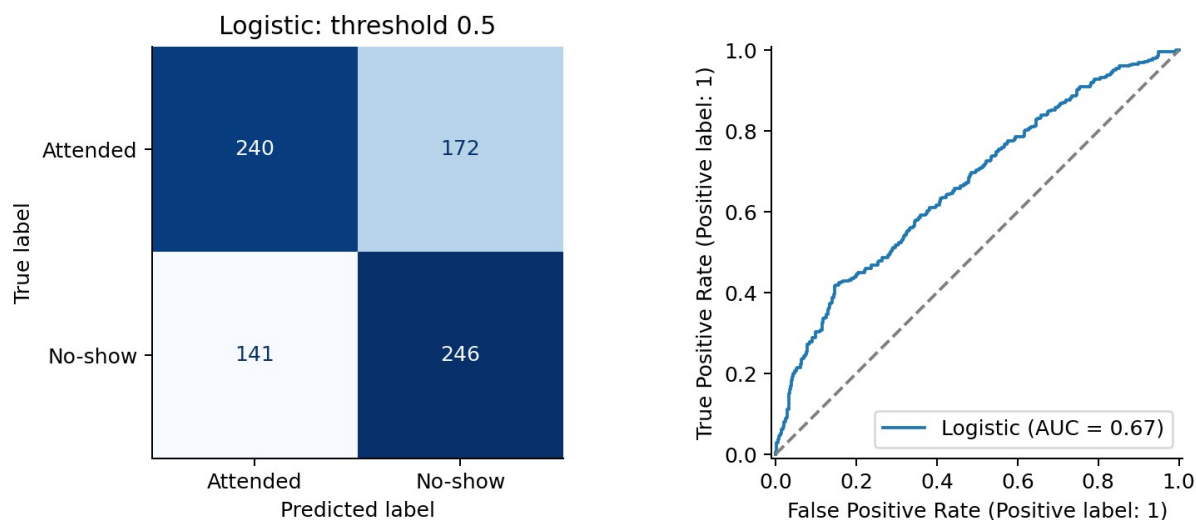
Metric	Dummy	Logistic regression
Accuracy	0.4844	0.6083
Precision	0.4844	0.5885
Recall	1.0000	0.6357
F1-score	0.6526	0.6112
ROC-AUC	0.5000	0.6678
Average precision	0.4844	0.6586
Brier score (lower better)	0.5156	0.2290

The dummy predicts no-show for all 799 test records because no-show is the training majority. Its recall is 1.0 and its F1 (0.6526) exceeds logistic regression (0.6112). I therefore do not claim that the trained model improves every metric. Logistic regression improves accuracy and ranking while reducing follow-up volume from 799 to 418 appointments.

The dummy uses hard probabilities of 1, which makes it a weak probability benchmark. A constant training-prevalence forecast would have a Brier score of approximately 0.2505; logistic regression scores 0.2290. This comparison is descriptive, not a confidence-tested superiority claim. Average precision is not the same numerical definition as trapezoidal PR-AUC.

A 0.5 probability threshold is retained as a technical reference. Choosing a service threshold requires staff capacity, costs and training-only validation. I did not choose a threshold by inspecting this test set.

7. What the errors mean for HealthConnect



At threshold 0.5, logistic regression produced 246 true positives, 240 true negatives, 172 false positives and 141 false negatives. It flagged 418 of 799 appointments (52.32%). If every flagged appointment received follow-up, 172 would be appointments that were actually attended, while 141 actual no-shows would not be flagged.

These numbers describe prediction errors, not measured intervention outcomes. Staff support might change attendance after deployment, and no intervention experiment has been conducted. High recall alone would create a large workload; the dummy demonstrates this by flagging everyone.

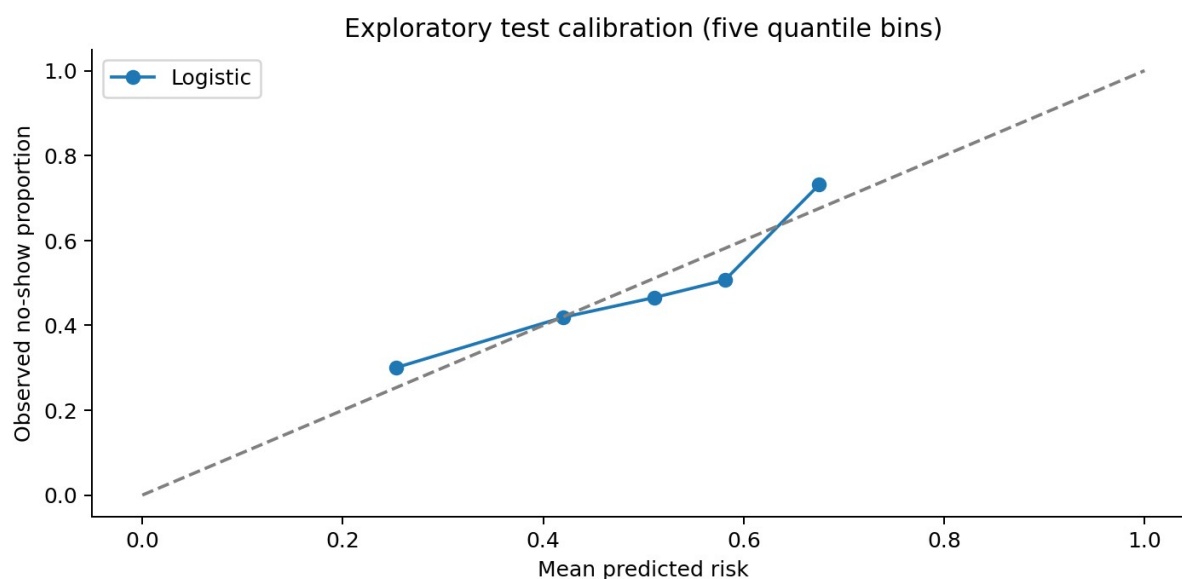


Figure 5. Five quantile bins compare mean predicted risk with observed test outcomes. Departures from the diagonal indicate calibration error. This small, descriptive check is not a calibration guarantee or reason to refit on the test set. Coefficients are exported for inspection but are conditional associations, not causal effects or directly comparable universal importance scores.

8. Risks, collaboration and Week 6 direction

I computed subgroup recall and false-positive rates by recorded gender and age group, retaining group sizes and positive counts in subgroup_metrics.csv. The audit is preliminary: small groups, inconsistent demographics and repeated patients limit inference. I do not claim fairness has been established, and I do not use row-independent confidence intervals to overstate certainty.

Resolved or mitigated

Parsing and raw-file preservation are verified. Missing distance is handled within training-only preprocessing. Excluding reminders and waiting time reduces unverified timing risk.

Still unresolved

Sunday conflicts, unreliable histories, unverified history timestamps, future cancellation treatment, synthetic-data realism and intervention capacity remain limitations. No real-world attendance benefit or production readiness is established.

Cross-track collaboration: pending

I prepared an ML Engineering handoff with the predictor schema, preprocessing requirements, timing assumptions and metrics. No colleague interaction or exchange has been evidenced. Before submission I must add the track/person, date, actual exchange, relevance and resulting change; the prepared handoff is not completed collaboration.

Week 6 priorities

First validate source history and timing. Then compare no-history and Sunday sensitivity analyses, a simple tree model and time-based training validation. Choose operating thresholds using validation and clinic workload. Keep a fresh holdout for later adaptations and evaluate cancellation handling before all-bookings deployment.

References and reproducibility

AnalystLab Africa Week 5 Experience Lab Assignment: Data Science tasks 1-6, collaboration, summary and submission sections. HealthConnect Week 4 ML Problem Definition Report: sections 3-9. HealthConnect Week 4 Project Summary. Supplied appointment CSV and data dictionary. IBM Data Science Methodology as studied in the course. Notebook and CSV outputs in this package are the numerical evidence. Exact environment versions and source digest are recorded in run_summary.json and requirements.txt.